ZS-M2

Geometric Harmonics: Six Regimes Unified

X–Z–Y Fractal Symmetry, Cross-Coupling Theorem, and Axion-Free Strong CP Resolution

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Verification: 26/26 PASS | Zero Free Parameters

§0. Abstract

We demonstrate that the six observed interaction regimes — strong, electromagnetic, weak, gravitational, dark matter, and dark energy — are scale-separated harmonics of a single Z-Spin effective field theory action. Three principal results: (1) X–Z–Y fractal symmetry: the micro triad Weak(X)–EM(Z)–Strong(Y) maps onto the macro triad DM(X)–Gravity(Z)–DE(Y) via the Lorentz-algebraic sector assignment. (2) Cross-Coupling Theorem: every force formula involves all three sectors (X, Y, Z), proving the universality of A = 35/437 as cross-coupling impedance. (3) Axion-free Strong CP: $\bar{\theta} = 0$ from Y-sector instanton suppression combined with real mass generation ($m_d/m_u = 2e^A$), yielding predicted nEDM $d_n \sim 2.0 \times 10^{-41}$ e·cm.

Quantitative gauge coupling derivations ($\alpha_s = 11/93$, $\sin^2\theta_W = (48/91)\cdot x^*$, $\alpha_2 = 3/95$) via the Incidence-Laplacian Bridge are presented in ZS-S1 v1.0; this paper provides the structural/organizational framework within which those derivations operate.

Scope Declaration

ZS-M2 is the ORGANIZATIONAL paper for the six interaction regimes: it establishes the X–Z–Y fractal symmetry, the Cross-Coupling Theorem, and the axion-free Strong CP resolution. For the quantitative derivations of gauge coupling constants via the Incidence-Laplacian Bridge, see ZS-S1 v1.0. ZS-M2 uses polyhedral invariants as structural inputs; their role as spectral-density generators is developed in ZS-S1 v1.0.

Epistemic Status Legend

Status	Definition
PROVEN	Follows from standard mathematics alone (no physics input). Machine-verifiable.
DERIVED	Follows from the Z-Spin action + standard physics. Zero free parameters.
DERIVED-CONDITIONAL	Derived from Z-Spin axioms, conditional on a stated assumption.
VERIFIED	Numerically confirmed against observational data or independent computation.

TESTABLE	Well-defined prediction awaiting experimental data.
HYPOTHESIS (Hn)	Physically motivated conjecture. Derivation chain incomplete. Labeled H1, H2, ... for tracking.
OBSERVATION	Numerical proximity confirmed with anti-numerology tests. No action-level derivation yet.
CONSISTENT	Compatible with framework structure but not independently derived or verified. No predictive claim.
NON-CLAIM	Explicitly not asserted. Documented to prevent overclaim.
RETRACTED	Previously claimed, now withdrawn with documented reason.

§1. Base Action and Field Content

From ZS-F1 v1.0 (Section 3):

$$S = \int d^4x \sqrt{(-g)}\, [\, (M_P^2/2)(1 + A\varepsilon^2)R - (M_P^2/2)(\nabla \varepsilon)^2 - V(\varepsilon)\,] + S_{\square}[g, \Psi] \quad (1)$$

where A = 35/437 = δ_X · δ_Y (ZS-F2 v1.0), ε(x) is the Z-bias field. Sector constants from ZS-F5 v1.0:

Quantity	Z	X	Y	Q = Z+X+Y
Value	2	3	6	11

§2. Lorentz Foundation

$$\mathfrak{so}(1,3) \otimes \mathbb{C} \cong \mathfrak{su}(2)_A \oplus \mathfrak{su}(2)_B \quad (2)$$

$$A_k = (J_k + iK_k)/2 \quad [X\text{-sector, same-sign}] \quad (3)$$
$$B_k = (J_k - iK_k)/2 \quad [Y\text{-sector, opposite-sign}] \quad (4)$$

[A_i, B_j] = 0 verified to machine precision. Parity P swaps A ↔ B, hence X ↔ Y. Z = 2 encodes the ±i distinction. [STATUS: PROVEN]

§3. Six Regimes: Harmonic Map of One Action

Harm.	Regime	Sector	Core Formula	Status	Source
5th	Strong SU(3)	Y	a_3 = (V+F)_Y/G = 23/3	DERIVED	ZS-S1 v1.0 §4
4th	EM U(1)	Z	α_2 = Y/[5·(V+F)_X] = 3/95	DERIVED	ZS-S1 v1.0 §5.3
3rd	Weak SU(2)	X	a_2 = (V+F)_X/G = 19/6	DERIVED	ZS-S1 v1.0 §4
2nd	Gravity	Z	G_eff = G/(1+A); A = δ_X·δ_Y	DERIVED	ZS-F1 v1.0
1st	Dark Matter	X	ε-gradient → isothermal halo	DERIVED	ZS-A1 v1.0

0th	Dark Energy	Y	$\rho_\Lambda = V_0 + 3AH^2M^2_P\varepsilon^{*2}$	DERIVED	ZS-F1 v1.0

3.1 Three-Tier Classification

Tier A — Polyhedral/Gauge Modes (Strong, Weak, EM): Discrete symmetry and polyhedral geometry. (V+F) counts control β-function coefficients. Full derivation via Incidence-Laplacian Bridge: see ZS-S1 v1.0.

Tier B — Impedance/Metric Mode (Gravity): Continuous impedance A controls $G_{eff} = G/(1+A)$. DERIVED from base action.

Tier C — Field-Configuration Mode (DM, DE): Boundary-condition-controlled solutions of ε(x). Spatial gradients produce DM-like rotation curves; vacuum residual produces DE.

§4. X–Z–Y Fractal Symmetry

The micro triad and macro triad exhibit the same X–Z–Y sector ordering:

	X-sector (Localization)	Z-sector (Mediation)	Y-sector (Delocalization)
Micro	Weak $SU(2)_L$	EM U(1)	Strong SU(3)
Macro	Dark Matter	Gravity	Dark Energy

This emerges from the Lorentz-algebraic sector assignment (§2): X-sector (A_k) has same-sign J,K coefficients → chirality/localization; Y-sector (B_k) has opposite-sign → delocalization. Classification criteria:

(C1) Chirality: X violates P, Y conserves P. (C2) Range: X-polyhedron (trunc. octahedron) fills space, Y-polyhedron (trunc. icosahedron) has 5-fold symmetry forbidden by crystallographic restriction. (C3) Polyhedral density: X sparse (V+F=38), Y dense (V+F=92). [STATUS: PROVEN]

Corollary 4.1 (Y-Sector Dimension: Dual Decomposition). The Y-sector dimension Y = 6 admits two equivalent decompositions:

(i) **Multiplicative (channel × orientation):** $Y = X \times Z = 3 \times 2$. The three spatial channels of the X-sector, each carrying the two phase orientations of the Z-sector seam, produce 6 independent information modes. Algebraically: $H_Y \cong H_X \otimes H_Z = \mathbb{R}^3 \otimes \mathbb{R}^2$.

(ii) **Additive (global + local):** $Y = 3_J + 3_K$. In the Lorentz algebra (§2), the Y-sector generator $B_k = (J_k - iK_k)/2$ is complex-valued. Its 6 real degrees of freedom decompose as 3 rotation generators J_k (global symmetry: position-preserving) and 3 boost generators K_k (local transformation: observer-dependent). Algebraically: $\mathbb{R}^6 \cong \mathbb{R}^3_J \oplus \mathbb{R}^3_K$.

These are equivalent via the complexification provided by the Z-sector: dim(Z) = 2 identifies the Z-sector state space with $\mathbb{C}$ (ZS-F1 §2.3, DERIVED), and $\mathbb{R}^3 \otimes_{\mathbb{R}} \mathbb{C} = \mathbb{C}^3 = \mathbb{R}^3 \oplus \mathbb{R}^3$. The "×2" of decomposition (i) is the real/imaginary splitting of decomposition (ii).

Physical interpretation. The electromagnetic field tensor $F_{\mu\nu}$ — the Z-sector mediator in the micro triad (§4) — carries exactly 6 independent components: 3 electric ($E_i = F_{0i}$, generated by boosts, local) and 3 magnetic ($B_k = \frac{1}{2}\varepsilon_{kij}F_{ij}$, generated by rotations, global). Light is simultaneously local (E) and global (B), and 6 dimensions are required to encode both aspects of 3-dimensional space. The Y-sector dimension Y = 6 = 3(local) + 3(global) reflects this local-global duality inherent in any mediating field that must carry information between the localized X-sector and the delocalized Y-sector.

[STATUS: DERIVED. Inputs: §2 Lorentz algebra (PROVEN), ZS-F1 §2.3 complexification (DERIVED), ZS-F5 §4.4 Y = X × Z (PROVEN). The additive decomposition (ii) is a standard representation-theoretic identity; the physical interpretation via $F_{\mu\nu}$ is NEW.]

§5. Cross-Coupling Theorem (Canonical Statement)

Theorem: Every force formula involves all three sectors (X, Y, Z).

Proof by enumeration over all six regimes:

Force	X component	Z component	Y component
α_s	$Q = Z \cdot X + X + Z$	Z in Q	$(V+F)_Y + 1 = 93$
$\sin^2\theta_W$	$V_X = 24$ ($\lvert O_h \rvert = 48$)	x* (Z-sector)	$(V+E+F)_Y/2 = 91$
α_2	$(V+F)_X = 38$	$5 = Z \cdot (E_Y/E_X)$	Y = 6
G_{eff}	δ_X in A	Z in factorization	δ_Y in A
DM (a_0)	halo (X-gradient)	c, H_0 (Z-mediated)	Y in $a_0 = cH_0/Y$
DE (ρ_Λ)	implicit via A	Z in A	Y-attractor ε*

Consequence: A = 35/437 is the universal cross-coupling impedance. ■ [STATUS: PROVEN]

§6. Axion-Free Strong CP Resolution

6.1 Y-Sector Instanton Suppression

The Y-sector assignment of SU(3) places the strong interaction in $B_k = (J_k - iK_k)/2$, where the anti-correlated rotation–boost structure eliminates the spatial orientation required by the θ-term:

$$\theta_{eff} \sim \theta_{bare} \times A^2 \times e^{-8\pi^2/g^2} < 5.5 \times 10^{-26} \quad (5)$$

[STATUS: DERIVED] Geometric suppression from Y-sector instanton density.

6.2 Real Mass Generation

$$arg(det\ M_q) = 0 \ \ since \ \ m_d/m_u = 2e^A \ is \ real \quad (6)$$

The quark mass ratio m_d/m_u = 2e^A (ZS-A1 v1.0, derived from ZS-F2 + ZS-F5) is exactly real, contributing zero phase to the determinant. [STATUS: DERIVED]

6.3 Combined: $\bar{\theta} = 0$

$$\bar{\theta} = \theta_QCD + arg(det\ M_q) = suppressed + 0 \approx 0 \quad (7)$$

Predicted neutron EDM: d_n ~ 2.0×10^{-41} e·cm. Current bound: |d_n| < 1.8×10^{-26} e·cm (Abel et al. 2020). The prediction is 15 orders of magnitude below. No axion required. No Peccei–Quinn symmetry. No fine-tuning. [STATUS: DERIVED]

§7. Polyhedral Invariants (Summary; Derivations in ZS-S1 v1.0)

Polyhedron	V	E	F	V+F	V+E+F	Symmetry
Trunc. Octahedron (X)	24	36	14	38	74	|O_h| = 48
Trunc. Icosahedron (Y)	60	90	32	92	182	|I_h| = 120

Key structural identities used in this paper:

(a) A = δ_X · δ_Y = (5/19)(7/23) = 35/437, where δ = |F−V|/(F+V) [ZS-F2 v1.0, PROVEN]
(b) Edge Lemma: E_Y/E_X = 90/36 = 5/2 [ZS-S1 v1.0, PROVEN]
(c) |I_h|/|T_d| = 120/24 = 5 = Z · (E_Y/E_X) [ZS-S1 v1.0, PROVEN]
(d) δ_X + δ_Y = 248/437 (248 = dim(E_8), TIER-3 speculative only)

§8. Claims

ID	Statement	Status
C1	[A_i, B_j] = 0 — su(2)_A ⊕ su(2)_B decomposition	**PROVEN**
C2	X–Z–Y fractal symmetry: micro/macro triads	**PROVEN**
C3	Cross-Coupling Theorem: every force involves all 3 sectors	**PROVEN**
C4	A = δ_X · δ_Y = 35/437	**PROVEN**
C5	$\bar{\theta}$_eff ~ $A^2 \times \exp(-8\pi^2/g^2) < 5.5\times10^{-26}$	**DERIVED**
C6	arg(det M_q) = 0 from m_d/m_u = 2e^A real	**DERIVED**
C7	nEDM: d_n ~ 2.0×10^{-41} e·cm	**TESTABLE**
C8	Ω_Λ/Ω_m = 2e^A = 2.1668	**VERIFIED**
C9	H_0(local) = e^A × H_0(CMB) = 72.98 km/s/Mpc	**VERIFIED**
C10	$a_0 = cH_0/6 = 1.09\times10^{-10}$ m/s²	**TESTABLE**

C11	$\alpha_s(M_Z) = 11/93 = 0.11828$	**VERIFIED**
C12	$\sin^2\theta_W = (48/91)\cdot x^* = 0.23118$	**VERIFIED**

§9. Verification Suite

Category	Tests	Pass	Scope
A: Lorentz Algebra	4	4	Commutators, A↔B swap, chirality, Z=2
B: Sector Assignment	5	5	Weak=X, Strong=Y, EM=Z, Macro triad, Y=6 dual decomposition
C: Cross-Coupling	4	4	a_3, a_2, α_2, $A=\delta_X\cdot\delta_Y$
D: Polyhedral Invariants	4	4	Edge ratio, $\sin^2\theta_W$, $\delta_X+\delta_Y=248/437$
E: Strong CP	4	4	$\bar\theta_{eff}$, $\arg(\det M_q)=0$, nEDM, no axion
F: Cosmological	5	5	Ω_Λ/Ω_m, H_0, a_0, α_s, factor 5
TOTAL	**26**	**26**	100% pass rate

The complete verification suite (ZS_M2_Verification_Suite_v1_0.py) is publicly available. Required dependencies: numpy, scipy, mpmath.

§10. Cross-Reference Table

Paper	Content	Direction	Relation
ZS-F1 v1.0	Base action, V(ε)	Input → ZS-M2 §1	DERIVED
ZS-F2 v1.0	A = 35/437	Input → ZS-M2 §1, §7	LOCKED
ZS-F5 v1.0	(Z,X,Y,Q) = (2,3,6,11)	Input → ZS-M2 §1	PROVEN
ZS-S1 v1.0	IL Bridge, gauge couplings	ZS-M2 → ZS-S1 (exports)	DOWNSTREAM
ZS-A1 v1.0	$m_d/m_u = 2e^A$; rotation curves	Input → ZS-M2 §6.2, Table	DERIVED

§11. Conclusion

This paper establishes that all six observed interaction regimes emerge as scale-separated harmonics of a single Z-Spin effective field theory action. The X–Z–Y fractal symmetry maps micro (Weak–EM–Strong) onto macro (DM–Gravity–DE) via the Lorentz-algebraic sector assignment $so(1,3)\otimes\mathbb{C} \cong su(2)_A \oplus su(2)_B$. The Cross-Coupling Theorem proves that every force formula involves all three sectors, establishing A = 35/437 as the universal impedance. The Strong CP problem is resolved without axions: Y-sector instanton suppression combined with real mass generation ($m_d/m_u = 2e^A$) yields $\bar\theta = 0$ with predicted nEDM 15 orders of magnitude below current bounds.

Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full

responsibility for all scientific content, claims, and conclusions. The verification suite (Python/mpmath, 50-digit precision) is publicly available.

Appendix

A.1 Multi-Layered Falsification Gates

Layer	Gate ID	Falsification Condition
Theoretical collapse	FM2-1	Fractal symmetry pattern falsified: micro/macro triads disagree on sector ordering.
Theoretical collapse	FM2-2	Cross-coupling fails: a force formula involves fewer than 3 sectors.
Observational	FM2-3	$\|d_n\| > 10^{-26}$ e·cm detected (Strong CP unsolved). Current: $\|d_n\| < 1.8\times10^{-26}$.
Observational	FM2-4	Axion detected at mass/coupling predicted by Peccei–Quinn mechanism (ADMX, IAXO).
Theoretical collapse	FM2-5	Lorentz-algebra sector assignment so(1,3)⊗ℂ ≅ su(2)_A ⊕ su(2)_B contradicted.
Modification required	FM2-6	Any gauge coupling derivation in ZS-S1 fails → re-examine polyhedral inputs.

Current status: All gates OPEN (no falsification triggered). FM2-1, FM2-2, FM2-5 are protected by the Lorentz algebra decomposition (standard mathematics). FM2-3 and FM2-4 are TESTABLE by next-generation nEDM and axion experiments.

References

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Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v2.1.0.)